# Hirmay Sandesara

Computational science, tensor networks, and quantum simulation

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## Research Interests

My primary interest is in researching and developing **compressed algorithms and frameworks** for quantum many-body simulation. I am interested in memory-efficient representations, regimes where **tensor-network ranks remain manageable**, and scalable methods for fermionic dynamics and finite-temperature simulation.

## Education

**Technical University of Munich** — Munich, Germany
*M.Sc. Computational Science and Engineering, quantum computing focus* — Oct 2024 – present

- Current grade: **1.5/5** (German scale; approximately 3.65/4.0).
- **TUM merit-based tuition waiver**; **Deutschlandstipendium**; CSE student representative.

**Ahmedabad University** — Ahmedabad, India
*B.Tech. Computer Science and Engineering* — Jul 2019 – May 2023

- Grade: 1.4 (German equivalent; 3.73/4.0); ranked **2nd of 140 students**.
- Dean's Recognition List; graduated with a certificate of highest excellence.

## Selected Publications

**H. Sandesara**, B. Decker, and C. Mendl. "MPS-Compressed Majorana Propagation with Virtual $\mathbb{Z}_2$ Charges." Manuscript in preparation.

**H. Sandesara**, A. Shukla, and P. Vedula. "A quantum approach for optimal control." *Quantum Information Processing* **24**, 95 (2025). doi:10.1007/s11128-025-04710-z.

**H. Sandesara** et al. "Space Sustainability Using Quantum Computing." White paper, OSF Preprints (2022). doi:10.31219/osf.io/wndfb.

## Research Experience

**Technical University of Munich** — Munich, Germany
*Master's thesis in tensor-network algorithms; supervisor: Prof. Christian B. Mendl* — Current

- Developing an **operator-space matrix product state (MPS)** approach for propagating Majorana operators in fermionic systems with long-range interactions.
- Introducing **virtual $\mathbb{Z}_2$ charge labels** on the MPS bonds to record accumulated fermionic parity, with the local rule $q_R = q_L \oplus p(s)$.
- Handling signs from long Jordan–Wigner and Majorana strings through endpoint and virtual-state information while preserving charge-aware tensor contractions and **SVD compression**.

**Fraunhofer Institute for Integrated Circuits IIS**
*Student Researcher, Quantum Compilation Group* — May 2026 – Oct 2026
Supervisors: Sarah Volkamer and Daniel D. Scherer.

- Integrated existing **automated cut-selection code** into a circuit-cutting software package under development, adapting the implementation for use within the package workflow.
- Surveyed compilation and cut-selection methods for quantum circuit cutting, including multi-qubit gate cutting, state-of-the-art approaches, open research questions, and relevant concepts from ZX calculus; prepared selected findings for internal technical discussions.
- Created Jupyter notebooks to demonstrate and test selected cut-finding approaches and their integration with the circuit-cutting package.

**Ahmedabad University** — Ahmedabad, India
*Research Assistant* — Sep 2023 – May 2024

- Developed a multidimensional formulation for nonlinear optimal-control problems using constrained Hamiltonian dynamics and canonical quantization.
- Designed and implemented a **variational quantum method** for the resulting broken-PT-symmetric, non-Hermitian systems using Python and Mathematica.
- Evaluated control problems with PennyLane, Qibo, SciPy, symbolic computation, parallel integration, and PDE solvers; work published in *Quantum Information Processing*.

**Quantum Open Source Foundation** Remote
*Mentee; mentor: Ryan Hill, qBraid* Apr 2024 – Aug 2024

- Selected as one of approximately 40 participants from more than 200 applicants.
- Worked on the design of a Python framework for configurable quantum reservoir-computing models and integration with QuEra computing resources.

**Artificial Brain** Remote
*Quantum Computing Intern* Jul 2022 – Dec 2022

- Co-authored a white paper formulating space-debris removal as knapsack and routing optimization problems and surveying QAOA and quantum-annealing approaches.
- Contributed to quantum scheduling for Earth-observation satellites and to a D-Wave portfolio-optimization prototype targeting the Sharpe ratio.

## Selected Technical Projects

**Entanglement distillation on a constrained quantum network** 2026
*MIT iQuHACK, IonQ Challenge – 2nd prize*

- Led a four-person team that designed a heuristic parity-check protocol for entanglement distillation under a minimal single-Bell-pair budget.
- Achieved a challenge score of 89 and claimed network edges at fidelity at least 0.9; see the code and technical report.

**NISQ methods for open-system simulation** 2024
*QHack Open Hackathon – 3rd prize, Seeing the Future; AWS Top 3*

- Led development of shallow parameterized circuits for state preparation and time evolution in open quantum systems.
- Tested the state-preparation routine on synthetic inputs and Alzheimer's patient data using **parallel GPU optimization** with NVIDIA cuQuantum; also explored logical cat-qubit simulations. Code and report.

**Hamiltonian simulation of coupled classical oscillators** 2024
*Classiq QRISE Challenge – winning submission*

- Derived the Hamiltonian and implemented state preparation and simulation workflows for a coupled-oscillator model on Classiq's platform.
- Compared construction choices with the practical aim of maintaining fidelity while reducing circuit depth. Project repository.

## Honors and Awards

**2nd Prize**, MIT iQuHACK IonQ Challenge 2026
**3rd Prize**, QHack Seeing the Future Challenge; AWS Top 3 2024
**Winner**, QRISE Classiq Challenge 2024
**TUM merit-based tuition waiver** and **Deutschlandstipendium** 2024
**Rank 2/140**, Ahmedabad University; Dean's Recognition List 2023

## Open Source, Mentoring, and Service

- Contributed an implementation of uniform-superposition state preparation to Qiskit's circuit library; co-authored material on Grover's algorithm for the Qiskit Textbook; and helped fix a Qiskit Terra bug.
- Translated more than 5,000 words of Qiskit educational material into Hindi.
- Mentored participants during the IBM Quantum Challenges in 2022 and 2024 by debugging tasks and explaining relevant concepts.
- Organized quantum-computing workshops and hackathons.

## Technical Skills

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|---|---|
| **Numerical methods** | Matrix product states/operators, tensor contractions, SVD and truncation, numerical linear algebra, optimization, PDE solvers, quantum simulation |
| **Programming** | Python, Julia, C, C++, MATLAB, Mathematica, SQL |
| **Libraries and tools** | NumPy, SciPy, SymPy, Qiskit, PennyLane, Qibo, QuTiP, cuQuantum, Matplotlib, PyTest, Git, LaTeX |